A-A-53903
30 May 1989

COMMERCIAL ITEM DESCRIPTION

STIMULATOR, ULTRASOUND, HIGH GALVANIC, HVS

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an ultrasound/high galvanic stimulator capable of independent or simultaneous application of high voltage stimulation or ultrasound.

Salient characteristics:

The ultrasound/high galvanic stimulator shall be capable of independent or simultaneous application of high voltage stimulation or ultrasound.

The stimulator shall have a peak current/voltage, and watts and watts per cm^2 readouts.

The stimulator shall have four operation modes: continuous, reciprocate, surge on-off, and probe.

The stimulator shall have intensity safety, positive or negative electrode polarity, pulse, and timer controls.

The stimulator shall have a head temperature monitor.

The stimulator shall have an ultrasound frequency of 1 MHz, plus or minus 5 percent, a crystal area of approximately 10 cm^2, an output power adjustable to 20 watts, and output intensity for high voltage stimulation of 500 volts maximum, and output intensity for ultrasound of 2.3 watts/cm^2 maximum, and an output current for high voltage stimulation of 2500 milliamps peak.

The stimulator shall have adjustable pulse from 1 to 200 pulses per second and output meter display in watts and watts per cm^2.

The ultrasound stimulator shall be provided with a treatment timer which shall be calibrated from 1 to 30 minutes.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Workmanship. The high galvanic ultrasound stimulator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One ultrasound stimulator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one high galvanic ultrasound stimulator as specified in this description.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Defects.

Stimulator does not power up on specified voltages and frequencies.

Stimulator mode selector switch does not select continuous, reciprocate, surge on-off, and probe modes.

Peak current/voltage, and watts and watts per cm^2 readouts do not function.

Intensity safety, positive or negative electrode polarity, pulse and timer controls do not function.

Stimulator head temperature monitor does not function.

Sampling for tests. Sampling for tests (dimensional) shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one high galvanic ultrasound stimulator.

a. First article. Where a first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. When a first article sample has not been required, one unit shall be subjected to the performance test specified herein. If a test article sample has successfully passed all tests, performance test shall not be required at acceptance.

Tests.

1. Plug into line monitor; line monitor should read 120 volts.

2. Turn timer on.

3. Put applicator into UPM-DT-10 ultrasound power meter; center over cone.

4. Re-zero U.P.M.

5. Turn intensity adj. to 20 watts.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:26Z
Check the source to verify that this is the current version before use.

Duty Cycle	In Tolerance Range
100 percent	17.2 - 22.8
50 percent	8.60 - 11.4
20 percent	3.44 - 4.56

6. Turn intensity to max.

7. Turn duty cycle from 20 percent to 50 percent to 100 percent. Pointer on meter should read 20-25W.

8. Turn function switch to combination mode. Ultrasound should reset and both pad LEDS on unit should be on.

9. Obtain brown test box.

10. Attach dispersive pad to unit and place on test box.

11. Set both intensity knobs to max, touch applicator to test box. Lamp on test box should be on. Volts/ammeter is 1000-1500 ma. Press voltage/current button and meter goes to 500V. Press again, meter goes back 1000-1500 ma. Turn intensity off.

12. Set function switch to continuous. Both meters should reset, with pad LEDS on. Hookup 4" red and black pads to active lead and to unit and place on test box. Turn stimulator intensity to max; meter should read between 2000-2500 ma. Press voltage/current switch; volts 500 plus or minus 10V. Push voltage/current switch to select current. Both pad lamps or test box should be on.

13. Switch polarity switch and note that when you change from plus to minus, lights on the test box switch polarity indication.

14. Set pulses sec. adj. to minimum, lights on the test box and green LED on the front panel should flicker. At 40 pulse sec. lamps and LEDS should be flashing very rapidly. Turn pulses/sec. back to max, LED and lamps should be solid.

15. Set function switch to surge and "surge on" and "surge off" to 2 sec. Reset intensity switch and adjust intensity to max; lights on test box and yellow LED on front panel

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:26Z
Check the source to verify that this is the current version before use.

should also go on and off. Observe current ramp on both test box and unit's meter to 2000-2500 ma. Voltage will stay at 500V and not ramp. Turn surge on and off down to 0.5 sec.; test box and unit surge lights will go on and off, meter will read 2000-2500 ma.

16. Set function switch to reciprocate 2.5 One pad LED at a time should be on. Reset intensity switch and adjust intensity to max. Meter will read between 1000-1500 ma. Voltage should read 500V. Observe that when red pad LED is on, the red pad is active on the test box. Repeat for black pad. Test box lights and unit lights will go from red to black.

17. Set function switch to reciprocate 5; meter will read between 1000-1500 ma; lights will go from red to black.

18. Set function switch to reciprocate 10, meter will read between 1000-1500 ma. Set intensity knob to 1000 ma. Lights will go from red to black. When black light comes on, turn black intensity knob fully ccw. Verify meter drops to 800-860 ma; turn black intensity knob back to max. When red lights come on, turn red intensity knob fully ccw. Verify meter drops to 800-860 ma; turn red intensity knob to max.

19. Set function switch to continuous, intensity off. Change pads to bifercated hookup. Turn intensity switch on to max intensity; verify lights are lighted on test box.

20. Turn unit timer to one minute; verify timer runs to zero and audible "ding" occurs.

21. Turn function switch on probe. Both pad LEDS should be off. Touch probe to test box, light should come on; adjust probe intensity control to max cw, volt-ammeter should read 1000-1500 ma; push voltage/current button, meter should read 500 volts. Push voltage current switch back to current and readjust probe intensity to minimum.

22. The stimulator source and sink risk currents shall be within safe limits as described in American National Standard, Safe Current Limits for Electromedical Apparatus, ANSI/AAMI ESl-1985. The test methods shall be as described in paragraph 4.3 and 4.4, ANSI/AAMI ESl-1985.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror contractor shall comply, and be responsible for compliance by its subcontractors, suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:26Z
Check the source to verify that this is the current version before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-01-284-6779

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4490

Location:
CID/STIMULATOR/H17-33

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.